

Leukemia Guide

A complete guide to the leukemias: what they are, the different variants, their causes, how to recognise them, how they are diagnosed, and how they are treated.

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1. Understanding Leukemia

Leukemia is a cancer of the blood-forming tissue, principally the bone marrow. The word comes from the Greek *leukos* (white) and *haima* (blood), coined in the mid-nineteenth century by Rudolf Virchow when he observed blood that appeared pale and thick because of an overwhelming excess of white cells. In modern terms, leukemia is a clonal malignancy: a single blood-forming stem or progenitor cell acquires genetic changes that give it a growth or survival advantage, and its descendants multiply until they crowd out the normal marrow.

Unlike most cancers, leukemia does not usually form a solid tumour. The malignant cells circulate in the blood and infiltrate the bone marrow, and frequently the spleen, liver, lymph nodes and other organs. The problems it causes therefore come from two directions at once: the malignant cells themselves (which may reach very high numbers, clog small vessels, or invade tissues) and the failure of normal blood production (which leads to anaemia, infection and bleeding).

Leukemia is not one disease. It is a family of related diseases distinguished by which lineage of blood cell is affected (myeloid or lymphoid), how mature the malignant cells are, and how quickly the disease progresses (acute or chronic). These distinctions matter enormously: they determine the symptoms a patient experiences, the tests needed, the treatment chosen and the likelihood of cure. A slow-growing chronic lymphocytic leukemia may need no treatment at all for many years, whereas an acute promyelocytic leukemia can be fatal within days if untreated but is now curable in the great majority of patients with the right therapy.

Leukemia accounts for roughly 3 percent of all new cancer diagnoses worldwide and is the commonest cancer of childhood, although about 90 percent of all leukemia cases occur in adults. Over the past four decades survival has improved dramatically for many subtypes, driven by better understanding of the genetics of the disease, refined risk-adapted chemotherapy, targeted drugs, immunotherapy and improvements in supportive care and stem cell transplantation.

2. How Blood Is Made and How Leukemia Disrupts It

Normal haematopoiesis

All blood cells descend from a small population of haematopoietic stem cells (HSCs) that reside in the bone marrow, mainly in the pelvis, vertebrae, sternum, ribs and the ends of long bones. Stem cells are able both to renew themselves and to differentiate. When they differentiate they commit to one of two great lineages:

- **Myeloid lineage** – gives rise to red blood cells (erythrocytes), platelets (via megakaryocytes), and the granulocytes (neutrophils, eosinophils, basophils) and monocytes that form the innate immune system.
- **Lymphoid lineage** – gives rise to B lymphocytes (which mature into antibody-producing plasma cells), T lymphocytes and natural killer (NK) cells, which constitute the adaptive immune system.

Differentiation proceeds through a series of progressively more committed progenitor cells. In the myeloid series, for example, a myeloblast matures through promyelocyte, myelocyte, metamyelocyte and band forms into a mature neutrophil. In the lymphoid series, a lymphoblast matures through several stages defined by surface proteins and gene rearrangements into a naive B or T cell. This entire process is tightly regulated by transcription factors (such as RUNX1, CEBPA, GATA1, PAX5, IKZF1), signalling pathways (FLT3, KIT, JAK-STAT, RAS) and epigenetic regulators (DNMT3A, TET2, IDH1/2, ASXL1).

A healthy adult produces on the order of 10^{11} to 10^{12} new blood cells every day. Mature cells released into the circulation have defined lifespans: red cells about 120 days, platelets about 7 to 10 days, neutrophils less than a day. The marrow therefore operates as a continuously running factory whose output is finely tuned to demand.

What goes wrong in leukemia

Leukemia arises when a stem cell or early progenitor acquires mutations that do two things: block normal maturation and drive uncontrolled proliferation or resistance to cell death. The classical model describes cooperating classes of mutation. Class I mutations (for example FLT3-ITD, KIT, NRAS, BCR::ABL1) activate signalling and confer a proliferative advantage. Class II mutations (for example RUNX1::RUNX1T1, CBFβ::MYH11, PML::RARA, NPM1) impair differentiation. Epigenetic and splicing mutations (DNMT3A, TET2, IDH1/2, ASXL1, SF3B1, SRSF2) alter the programme of gene expression more broadly and frequently appear early, sometimes years before overt disease.

The consequence in acute leukemia is the accumulation of immature cells called blasts, which are frozen at an early stage of development and unable to perform any useful function. In chronic leukemia the malignant cells retain a greater ability to mature, so the marrow and blood fill with cells that look relatively normal but are produced in excess and are functionally abnormal.

Because the marrow has limited space, the expanding leukemic clone displaces normal haematopoiesis. This produces the triad of marrow failure: anaemia (too few red cells), neutropenia (too few functioning white cells) and thrombocytopenia (too few platelets). Most of the symptoms and early complications of leukemia follow directly from this triad.

Clonal haematopoiesis: the precursor state

With sensitive sequencing, small clones of blood cells bearing leukemia-associated mutations (most commonly DNMT3A, TET2 and ASXL1) can be found in a substantial proportion of healthy older people. This is called clonal haematopoiesis of indeterminate potential (CHIP). CHIP is not leukemia, and the majority of people who carry it never develop leukemia, but it raises the annual risk of progression to a myeloid malignancy by roughly 0.5 to 1 percent per year and is also associated with cardiovascular disease. It illustrates that leukemia develops in steps over time rather than appearing fully formed.

3. Classification: The Four Major Types and Beyond

Leukemias are classified along two axes. The first is tempo: **acute** leukemias are composed of immature blasts, progress over days to weeks and are fatal within months if untreated; **chronic** leukemias are composed of more mature cells, progress over months to years and may be indolent for long periods. The second axis is lineage: **myeloid** or **lymphoid**. Combining the two gives the four principal types that account for the great majority of cases.

Type	Lineage	Tempo	Typical age	Approximate share of adult leukemia
Acute lymphoblastic leukemia (ALL)	Lymphoid (B or T)	Acute	Peak 2–5 years; second rise in older adults	About 10–15% (but about 75–80% of childhood leukemia)
Acute myeloid leukemia (AML)	Myeloid	Acute	Median about 68 years; incidence rises with age	About 30%

Type	Lineage	Tempo	Typical age	Approximate share of adult leukemia
Chronic lymphocytic leukemia (CLL)	Lymphoid (mature B)	Chronic	Median about 70 years; rare under 40	About 30–35% in Western populations; less common in Asia
Chronic myeloid leukemia (CML)	Myeloid	Chronic (with potential to transform)	Median about 55–65 years	About 15%

Beyond these four, there are a number of less common entities: acute promyelocytic leukemia (a distinctive subtype of AML with unique biology and treatment), hairy cell leukemia, prolymphocytic leukemias, large granular lymphocytic leukemia, chronic myelomonocytic leukemia, juvenile myelomonocytic leukemia, and mixed phenotype acute leukemia. Each is discussed later in this guide.

Formal classification systems

Historically the French–American–British (FAB) system classified acute leukemias by morphology and cytochemistry into AML subtypes M0 to M7 and ALL subtypes L1 to L3. FAB terms are still heard in practice (for example "M3" for promyelocytic leukemia), but they have been superseded. Current classification uses the World Health Organization (WHO) Classification of Haematolymphoid Tumours, 5th edition (2022), and the parallel International Consensus Classification (ICC, 2022). Both systems place genetic abnormalities at the centre of diagnosis, defining many entities by their driving translocation or mutation rather than by appearance alone. For AML, the WHO 2022 classification removed the 20 percent blast threshold for most genetically defined subtypes: for example, an NPM1-mutated or PML::RARA-positive leukemia is AML regardless of blast count.

Chronic leukemias are grouped with related conditions. CML sits within the myeloproliferative neoplasms; CLL sits within the mature B-cell neoplasms and is considered the same disease as small lymphocytic lymphoma (SLL), differing only in whether the malignant cells are predominantly in the blood or in lymph nodes.

4. Causes and Risk Factors

For the majority of individual patients, no specific cause of leukemia can be identified. Leukemia results from the accumulation of somatic mutations in blood-forming cells, and most of these mutations arise from random errors during DNA replication, a process that continues throughout life and accelerates with age. Nevertheless, a number of factors are known to raise the probability that such mutations occur or that a mutated clone survives and expands.

Age

Age is the strongest risk factor for the adult leukemias. The incidence of AML, CLL and CML rises steadily after the age of 50 and peaks in the seventh and eighth decades. ALL is the exception, with a sharp peak between two and five years of age; this early peak is believed to reflect a two-step process in which a prenatal initiating lesion (often a chromosomal translocation acquired in utero) is followed by a postnatal second hit, possibly triggered by an abnormal immune response to delayed infection exposure.

Ionising radiation

Radiation is the best-established environmental cause. Survivors of the atomic bombings of Hiroshima and Nagasaki showed a marked excess of leukemia, especially CML and AML, peaking five to seven

years after exposure and falling thereafter. Therapeutic radiation for other cancers and, historically, radiation used for benign conditions such as ankylosing spondylitis have also raised leukemia risk. Diagnostic imaging involves far smaller doses; the risk from a single CT scan is very small, but cumulative exposure from repeated scans, especially in childhood, is associated with a measurable increase.

Chemical exposures

Benzene is the most clearly established chemical leukemogen. Chronic occupational exposure (in the rubber, petrochemical, shoe and paint industries, and in the past through gasoline handling) is associated with AML and myelodysplastic syndromes. Benzene is also a constituent of cigarette smoke, and smoking is estimated to account for about 20 percent of adult AML cases in some populations. Formaldehyde, some pesticides and herbicides, and certain solvents have been linked to leukemia in epidemiological studies with weaker or inconsistent evidence.

Previous cancer treatment (therapy-related myeloid neoplasms)

Chemotherapy and radiotherapy given for an earlier cancer can damage the DNA of marrow stem cells and lead to a second, treatment-induced leukemia. Two patterns are recognised. Alkylating agents (cyclophosphamide, melphalan, chlorambucil, busulfan, platinum drugs) and radiation typically cause a myelodysplastic phase followed by AML, appearing five to ten years after exposure and frequently associated with loss of chromosomes 5 and 7 and TP53 mutation. Topoisomerase II inhibitors (etoposide, anthracyclines) cause an earlier-onset AML, often within one to three years, without a preceding dysplastic phase and typically with rearrangements of the KMT2A gene on chromosome 11q23. Therapy-related myeloid neoplasms account for roughly 10 percent of AML and carry a poor prognosis.

Inherited and constitutional conditions

Several inherited syndromes substantially increase leukemia risk. Children with Down syndrome (trisomy 21) have a 10- to 20-fold increased risk of both ALL and AML, and a distinctive transient abnormal myelopoiesis in the newborn period that can evolve into a specific form of AML with GATA1 mutation. Fanconi anaemia, Bloom syndrome, ataxia–telangiectasia, dyskeratosis congenita, Shwachman–Diamond syndrome, Diamond–Blackfan anaemia and neurofibromatosis type 1 all carry increased risk through defective DNA repair, telomere maintenance or ribosome function. Li–Fraumeni syndrome (germline TP53 mutation) increases the risk of many cancers including leukemia.

Increasingly, germline predisposition is recognised in families without a classical syndrome. Germline mutations in RUNX1, CEBPA, GATA2, DDX41, ETV6 and ANKRD26 are now recognised in the WHO classification as defining specific myeloid neoplasms with germline predisposition. Identifying these matters both for the patient and for relatives, and particularly when a sibling is being considered as a stem cell donor.

Pre-existing blood disorders

Myelodysplastic syndromes (MDS) and myeloproliferative neoplasms (polycythaemia vera, essential thrombocythaemia, primary myelofibrosis) are clonal marrow disorders that can evolve into AML. Between 10 and 30 percent of MDS patients progress to AML, and such secondary AML responds less well to standard treatment. Aplastic anaemia treated with immunosuppression and paroxysmal nocturnal haemoglobinuria are also associated with later myeloid malignancy.

Viruses

Only one leukemia has a proven viral cause: adult T-cell leukemia/lymphoma is caused by human T-lymphotropic virus type 1 (HTLV-1), endemic in southern Japan, the Caribbean, parts of South America, central Africa and Australia's Indigenous population. Infection is lifelong and about 5 percent of carriers develop the disease after a latency of decades. Epstein–Barr virus is associated with some lymphomas rather than leukemia as such, though it contributes to post-transplant lymphoproliferative disease.

Other factors

Obesity is associated with a modest increase in leukemia incidence. Occupational exposure to electromagnetic fields, living near power lines, hair dyes and mobile telephone use have all been studied extensively and the evidence for a meaningful effect is weak or absent. Leukemia is not contagious and cannot be passed from one person to another. A family history of CLL does raise an individual's risk severalfold, reflecting inherited susceptibility variants rather than a single gene, but the absolute risk remains low.

5. Recognising Leukemia: Signs and Symptoms

The presentation of leukemia is remarkably varied. Some patients are critically unwell within days; others are diagnosed incidentally from a blood test taken for another reason and feel entirely well. Nevertheless the manifestations follow logically from the underlying pathology, and can be grouped into those caused by marrow failure, those caused by the leukemic cells themselves, and constitutional symptoms.

Symptoms of bone marrow failure

Anaemia

A low red cell count is almost universal in acute leukemia and common in advanced chronic leukemia. Patients describe fatigue out of proportion to activity, breathlessness on exertion, light-headedness, palpitations, difficulty concentrating and, in older patients or those with coronary disease, chest pain. On examination there is pallor of the skin, conjunctivae, nail beds and palmar creases. Because the anaemia often develops over weeks, some patients adapt and under-report their symptoms.

Neutropenia and infection

A shortage of functional neutrophils leaves the patient vulnerable to bacterial and fungal infection. Recurrent or persistent infections (sore throat, mouth ulcers, gum infection, pneumonia, skin infections, perianal abscess) that fail to respond to the usual antibiotics should prompt a blood count. Fever in a patient with acute leukemia is a medical emergency, because without neutrophils the normal signs of infection may be absent and sepsis can develop within hours.

Thrombocytopenia and bleeding

Low platelets cause easy bruising, petechiae (pinpoint red or purple spots that do not blanch, typically over the shins and in dependent areas), purpura, nosebleeds, bleeding gums when brushing teeth, heavy menstrual bleeding and prolonged bleeding after minor cuts. In severe cases there may be gastrointestinal bleeding, blood in the urine, retinal haemorrhage or intracranial haemorrhage. In acute promyelocytic leukemia, bleeding is compounded by disseminated intravascular coagulation and may be the presenting and life-threatening feature.

Symptoms caused by the leukemic cells

Organ infiltration

Leukemic cells accumulate in the spleen and liver, producing splenomegaly and hepatomegaly, which may be felt as a dragging discomfort or fullness in the left upper abdomen, early satiety and weight loss. Massive splenomegaly is characteristic of CML and hairy cell leukemia. Lymph node enlargement (lymphadenopathy) is prominent in CLL and in ALL, particularly T-cell ALL, which may also present with a mediastinal mass causing cough, breathlessness, superior vena cava obstruction or stridor.

Bone and joint pain is common in childhood ALL, where marrow expansion stretches the periosteum; a child may limp, refuse to walk or be misdiagnosed with juvenile arthritis. Gum hypertrophy and skin infiltrates (leukemia cutis) are typical of monocytic subtypes of AML. Testicular enlargement can occur in boys with ALL. Solid collections of myeloid blasts outside the marrow, called myeloid sarcomas or chloromas, may appear in bone, skin, orbit or soft tissue, sometimes before marrow involvement is evident.

Central nervous system involvement

Leukemic cells may cross into the cerebrospinal fluid, especially in ALL and monocytic AML. This produces headache, vomiting, visual disturbance, cranial nerve palsies (commonly of the facial nerve or the nerves controlling eye movement), neck stiffness or, rarely, seizures. CNS involvement is present at diagnosis in about 5 percent of ALL cases and is a site of relapse without specific prophylaxis.

Leukostasis

When the white cell count is very high (typically above $100 \times 10^9/L$ in AML, higher in ALL and CML), the sheer volume of blasts increases blood viscosity and obstructs small vessels in the lungs, brain and retina. Patients develop breathlessness, hypoxia, confusion, headache, blurred vision and priapism. Leukostasis is an emergency requiring urgent cytoreduction.

Constitutional symptoms

Fever without infection, drenching night sweats and unintentional weight loss of more than 10 percent of body weight over six months are known collectively as B symptoms. They result from cytokine release by the malignant clone and are especially characteristic of the chronic leukemias and lymphoma-like presentations. Generalised malaise and loss of appetite are common.

Metabolic complications

Rapid turnover of leukemic cells releases uric acid, potassium and phosphate into the blood, and this is exaggerated when treatment begins. Tumour lysis syndrome causes hyperuricaemia, hyperkalaemia, hyperphosphataemia, hypocalcaemia and acute kidney injury. Hypercalcaemia is occasionally seen in adult T-cell leukemia.

Distinguishing patterns

Feature	Acute leukemia (ALL, AML)	Chronic leukemia (CLL, CML)
Onset	Days to weeks	Months to years; often incidental
Dominant symptoms	Marrow failure: fatigue, infection, bleeding	Often none; later fatigue, splenomegaly, lymphadenopathy, B symptoms
White cell count	Low, normal or very high	Usually high, sometimes markedly

Feature	Acute leukemia (ALL, AML)	Chronic leukemia (CLL, CML)
Blood film	Blasts present	Mature-looking lymphocytes (CLL) or full spectrum of myeloid cells (CML)
Anaemia and thrombocytopenia	Usually present and often severe	Absent early; develop with progression
Organomegaly	Variable; moderate	Splenomegaly common (may be massive in CML); nodes prominent in CLL
Urgency	Emergency; treatment within days	Elective work-up; CLL may be observed for years

Red flags that warrant a same-day full blood count: unexplained bruising or petechiae, persistent fever with no clear source, pallor and profound fatigue, bone pain in a child, a very high or very low white cell count on any blood test, or any combination of infection, bleeding and anaemia.

6. Diagnosis and Testing

The diagnosis of leukemia proceeds in stages: a screening blood count raises the suspicion; examination of the blood film narrows the possibilities; bone marrow examination confirms the diagnosis and provides material for specialised testing; and immunophenotyping, cytogenetics and molecular genetics define the precise subtype and risk category that will guide treatment. In current practice the last of these steps is indispensable, because the choice of therapy for many leukemias now depends on specific mutations or fusion genes.

Full blood count and blood film

The full blood count (FBC or CBC) measures haemoglobin, white cell count with differential, platelet count and red cell indices. In acute leukemia the typical picture is anaemia and thrombocytopenia with a white cell count that may be raised, normal or reduced; a normal count does not exclude leukemia. In CLL there is a persistent lymphocytosis (clonal B cells above $5 \times 10^9/L$ are required for the diagnosis). In CML there is marked leucocytosis, often above $100 \times 10^9/L$, with basophilia and frequently thrombocytosis.

The peripheral blood film, examined under the microscope by a trained haematologist or scientist, remains one of the most informative tests in medicine. Blasts are large cells with a high nuclear-to-cytoplasmic ratio, fine open chromatin and prominent nucleoli. Auer rods, needle-like crystalline inclusions in the cytoplasm, are pathognomonic of myeloid lineage. In CLL the film shows small mature lymphocytes and characteristic smear (smudge) cells that rupture during preparation. In CML there is a left shift with every stage of granulocyte maturation present, resembling a bone marrow in the blood. In hairy cell leukemia the cells have fine cytoplasmic projections. A leukoerythroblastic film, with nucleated red cells and immature granulocytes, indicates marrow infiltration or fibrosis.

Bone marrow aspirate and trephine biopsy

Bone marrow sampling is performed under local anaesthetic, usually from the posterior iliac crest, with the patient lying on their side. Sedation may be offered, and general anaesthesia is used in children. Two samples are taken: an aspirate, in which liquid marrow is drawn into a syringe and spread on slides for cytology, and a trephine (core) biopsy, in which a small cylinder of bone and marrow is removed for histology. The aspirate provides the blast percentage (20 percent or more of nucleated cells defines acute leukemia in the classical definition, although genetically defined subtypes may be diagnosed at lower counts), morphology and material for flow cytometry, cytogenetics and molecular testing. The

trephine shows cellularity, architecture, fibrosis and the pattern of infiltration, and is essential when the aspirate is dry, as happens in hairy cell leukemia and myelofibrosis.

Cytochemistry

Special stains applied to marrow smears help assign lineage when morphology is ambiguous. Myeloperoxidase (MPO) and Sudan black B stain myeloid blasts; non-specific esterase stains monocytic cells; periodic acid–Schiff (PAS) shows block positivity in lymphoblasts. Tartrate-resistant acid phosphatase (TRAP) was historically used for hairy cell leukemia. Cytochemistry has been largely supplanted by flow cytometry but remains useful in resource-limited settings.

Immunophenotyping by flow cytometry

Flow cytometry passes cells one by one through a laser and measures fluorescently labelled antibodies bound to surface and intracellular proteins (cluster of differentiation, or CD, antigens). It defines lineage and stage of maturation, distinguishes leukemia from reactive processes, and provides the baseline pattern used later to detect measurable residual disease. Key markers include:

Leukemia	Characteristic immunophenotype
B-lymphoblastic leukemia (B-ALL)	CD19, CD22, CD79a, CD10 (common ALL antigen; absent in pro-B), TdT, CD34 variable; surface immunoglobulin negative (positive in mature B-ALL / Burkitt)
T-lymphoblastic leukemia (T-ALL)	Cytoplasmic CD3, CD7, CD2, CD5, CD1a, TdT; CD4/CD8 co-expression common; early T-precursor (ETP) subtype lacks CD1a/CD8 and expresses myeloid/stem markers
Acute myeloid leukemia (AML)	CD13, CD33, CD117, MPO, CD34, HLA-DR; CD14/CD64/CD11b in monocytic subtypes; CD41/CD61 in megakaryoblastic; glycophorin A in erythroid
Acute promyelocytic leukemia (APL)	CD33 bright, CD13, MPO; typically CD34 negative and HLA-DR negative
Chronic lymphocytic leukemia (CLL)	CD5, CD19, CD23, CD200 positive; CD20 and surface immunoglobulin weak; FMC7 negative; light chain restricted
Hairy cell leukemia	CD19, CD20 bright, CD11c, CD25, CD103, CD123, annexin A1, BRAF V600E
Mixed phenotype acute leukemia	Co-expression of definitive markers of two lineages (e.g. MPO plus CD19/CD79a/CD22, or MPO plus cytoplasmic CD3)

Cytogenetics

Conventional karyotyping cultures marrow cells, arrests them in metaphase and examines the banded chromosomes under the microscope. It detects numerical changes (gains and losses of whole chromosomes such as trisomy 8, monosomy 7, hyperdiploidy in childhood ALL) and structural rearrangements (translocations, inversions, deletions). Fluorescence in situ hybridisation (FISH) uses labelled DNA probes to detect specific abnormalities rapidly in non-dividing cells and is used, for example, to confirm BCR::ABL1 in CML, PML::RARA in APL, and del(17p), del(11q), trisomy 12 and del(13q) in CLL. Cytogenetic findings are among the strongest prognostic markers in AML and ALL.

Molecular genetics

Polymerase chain reaction (PCR) and next-generation sequencing (NGS) identify mutations and fusion transcripts that may be invisible on karyotype. A targeted myeloid or lymphoid gene panel is now standard at diagnosis. The results serve three purposes: they establish the WHO diagnosis (for example NPM1-mutated AML), they stratify risk (for example FLT3-ITD in AML, IKZF1 deletion in ALL, TP53

mutation or unmutated IGHV in CLL), and they identify targets for therapy (FLT3, IDH1, IDH2, BCR::ABL1, BRAF, KMT2A, menin-dependent NPM1). Quantitative PCR for BCR::ABL1 transcripts is the backbone of CML monitoring, reported on the International Scale (IS) as a percentage relative to a standardised baseline.

Measurable residual disease (MRD)

MRD refers to leukemic cells that persist after treatment at levels below the sensitivity of morphology (which cannot reliably detect fewer than about 5 percent blasts). Multiparameter flow cytometry detects one leukemic cell in 10^4 to 10^5 normal cells; quantitative PCR for fusion genes, NPM1 or immunoglobulin/T-cell receptor rearrangements and NGS-based assays reach 10^{-5} to 10^{-6} . MRD status after induction and consolidation is the single most powerful predictor of relapse in ALL and in many forms of AML, and is now used to decide whether to escalate treatment (for example proceeding to transplantation) or de-escalate it. In CLL, undetectable MRD at the end of fixed-duration therapy predicts long treatment-free remission.

Lumbar puncture

Examination of the cerebrospinal fluid is mandatory at diagnosis in ALL and in AML with monocytic features, very high white counts or neurological symptoms. It is combined with the first dose of intrathecal chemotherapy. The fluid is examined by cytopspin and flow cytometry; CNS involvement is graded by the cell count and presence of blasts (CNS1, CNS2, CNS3).

Other investigations

- **Coagulation screen** (PT, APTT, fibrinogen, D-dimer) – essential at presentation, particularly to detect disseminated intravascular coagulation in APL and monocytic AML.
- **Biochemistry** – urea, creatinine, electrolytes, uric acid, phosphate, calcium and lactate dehydrogenase, to assess tumour lysis risk and organ function; liver function tests.
- **Virology** – hepatitis B and C, HIV, CMV, HTLV-1 where relevant, before immunosuppressive therapy and transplantation.
- **HLA typing** – performed at diagnosis in all patients who might be candidates for allogeneic transplant, so that a donor search can begin early.
- **Imaging** – chest radiograph or CT for mediastinal mass in T-ALL; CT or ultrasound for organomegaly and nodal disease in CLL; MRI for suspected CNS or bone involvement; PET-CT is not routine in leukemia but may be used in suspected Richter transformation of CLL.
- **Cardiac assessment** – echocardiogram or MUGA scan before anthracycline chemotherapy.
- **Fertility counselling** – sperm banking or, where time permits, oocyte or ovarian tissue cryopreservation before intensive therapy.
- **Germline testing** – considered when there is a family history, a young patient, or a mutation (such as RUNX1, DDX41, CEBPA, GATA2) that may be inherited.

Staging and risk stratification

Acute leukemias are not staged in the conventional sense; instead patients are stratified into risk groups (favourable, intermediate, adverse) based on age, white count, cytogenetics, molecular findings and MRD response. CLL is staged by the Rai system (0 to IV, used mainly in North America) or the Binet system (A to C, used in Europe), both of which combine blood counts with the number of involved lymphoid areas; the CLL International Prognostic Index (CLL-IPI) adds age, TP53 status, IGHV mutation status and beta-2 microglobulin. CML is staged by phase: chronic, accelerated and blast phase, defined

by blast percentage, basophilia, platelet count, spleen size and cytogenetic evolution.

7. Acute Lymphoblastic Leukemia (ALL)

Overview and epidemiology

Acute lymphoblastic leukemia is a malignancy of immature lymphoid precursors (lymphoblasts) committed to the B-cell lineage in about 85 percent of cases or the T-cell lineage in about 15 percent. It is the commonest childhood cancer, with an incidence peak between the ages of two and five, and cure rates in children now exceed 90 percent in high-income countries. A second, smaller incidence peak occurs after the age of 50. In adults the disease is biologically more adverse and outcomes historically poorer, although immunotherapy has transformed the picture over the past decade. Males are affected slightly more often than females.

Biology and subtypes

ALL is heterogeneous at the genetic level, and its subtypes have distinct prognoses. The WHO 2022 classification recognises B-lymphoblastic leukemia/lymphoma with the following recurrent genetic abnormalities, among others:

Subtype	Frequency and features	Prognosis
High hyperdiploidy (51–65 chromosomes)	About 25–30% of childhood B-ALL; gains of chromosomes 4, 10, 17, 21	Very favourable
ETV6::RUNX1, t(12;21)	About 25% of childhood B-ALL; cryptic on karyotype, detected by FISH/PCR	Very favourable
BCR::ABL1, t(9;22) (Philadelphia chromosome)	About 3% of children, 25% of adults, over 50% of adults above 60	Historically adverse; now much improved with tyrosine kinase inhibitors
BCR::ABL1-like (Ph-like)	About 15% of childhood, 20–25% of adult B-ALL; CRLF2, ABL-class, JAK2 rearrangements; IKZF1 deletions common	Adverse; targetable in some cases (ABL-class with TKIs, JAK pathway with ruxolitinib under study)
KMT2A-rearranged, t(4;11) and others	About 70% of infant ALL; 5–10% of adult ALL	Adverse, particularly in infants
TCF3::PBX1, t(1;19)	About 5%	Intermediate; improved with intensive therapy
Hypodiploidy (<44 chromosomes)	About 1–2%; near-haploid often has RAS mutations; low hypodiploid often TP53 (may be germline)	Adverse
iAMP21	About 2% of childhood B-ALL	Adverse with standard therapy; improved with intensification
DUX4, ZNF384, MEF2D, PAX5 alterations	Newer entities defined by RNA sequencing	DUX4 favourable; MEF2D adverse; others intermediate
T-ALL	15% of childhood, 25% of adult ALL; NOTCH1 mutations in >50%; ETP-ALL is a stem-like subtype	Intermediate; ETP historically adverse but improving with MRD-directed therapy

Clinical presentation

Most patients present with a short history of days to a few weeks. Children commonly show fatigue, pallor, fever, bruising, bone or joint pain and limping. Lymphadenopathy, hepatosplenomegaly and, in T-ALL, a mediastinal mass with airway compression or pleural effusion are frequent. CNS symptoms are present at diagnosis in a minority. In adults the pattern is similar, though B symptoms and organomegaly may be more prominent. Testicular involvement occurs in about 2 percent of boys.

Risk stratification

Risk assignment in childhood ALL uses the National Cancer Institute criteria (standard risk: age 1 to less than 10 years and white count under $50 \times 10^9/L$; high risk: otherwise), combined with cytogenetics, CNS status and, decisively, MRD at the end of induction (day 29) and consolidation. In adults, age over 35 to 40, high white count (over $30 \times 10^9/L$ in B-ALL, over $100 \times 10^9/L$ in T-ALL), adverse genetics and persistent MRD define high risk.

Treatment of ALL

Treatment for ALL is a prolonged, multi-phase programme lasting two to three years. The framework was developed in paediatric cooperative group trials and has been adapted, with modification, for adults. The phases are:

Induction (4 to 6 weeks)

The aim is to eliminate more than 99 percent of leukemic cells and restore normal haematopoiesis. Standard induction combines a glucocorticoid (prednisolone or dexamethasone), vincristine, an asparaginase preparation (pegaspargase or, where hypersensitivity develops, Erwinia asparaginase) and, in higher-risk or adult regimens, an anthracycline (daunorubicin or doxorubicin). Intrathecal chemotherapy is given from the outset. Complete remission (fewer than 5 percent marrow blasts with count recovery) is achieved in 95 to 98 percent of children and 85 to 90 percent of adults. In Philadelphia-positive ALL a tyrosine kinase inhibitor (imatinib, dasatinib or ponatinib) is added from induction onward, and combination with the bispecific antibody blinatumomab is increasingly used to reduce or replace chemotherapy.

Consolidation or intensification (several months)

Once remission is achieved, blocks of intensive chemotherapy with different agents are given to eradicate residual disease and prevent resistance. Commonly used drugs include cyclophosphamide, cytarabine, mercaptopurine, thioguanine, methotrexate (including high-dose methotrexate with leucovorin rescue), etoposide and further asparaginase. A delayed intensification phase repeating an induction-like block improves outcomes in children. For adults with B-ALL, the addition of blinatumomab during consolidation improved overall survival in the ECOG E1910 trial even in patients who were already MRD-negative, and it is now part of standard care in many regions. Rituximab is added for CD20-positive B-ALL in adults.

CNS-directed therapy

Because systemic chemotherapy penetrates the CNS poorly, all patients receive intrathecal methotrexate (with or without cytarabine and hydrocortisone) throughout treatment, plus high-dose systemic methotrexate and dexamethasone, which cross the blood–brain barrier. Cranial irradiation, once routine, is now reserved for overt CNS disease at diagnosis or relapse, because of its long-term effects on cognition, growth and secondary cancers.

Maintenance (18 to 30 months)

Low-intensity oral chemotherapy with daily mercaptopurine and weekly methotrexate, with periodic pulses of vincristine and glucocorticoid, continues for about two years in girls and adults and up to three

years in boys in some protocols. Adherence is critical; measurement of thiopurine metabolites and TPMT/NUDT15 genotyping guide dosing. Maintenance is essential in ALL and its omission substantially increases relapse.

Allogeneic stem cell transplantation

Transplantation in first remission is reserved for patients at high risk of relapse: persistent MRD after consolidation, adverse genetics such as KMT2A rearrangement or hypodiploidy, or induction failure. For Philadelphia-positive ALL, transplantation was once universal but is increasingly deferred in patients who achieve deep molecular remission on TKI-based therapy. Transplantation is the standard consolidation for most patients in second remission.

Immunotherapy

Three classes of immunotherapy have transformed the treatment of relapsed and refractory B-ALL and are moving into front-line use. **Blinatumomab** is a bispecific T-cell engager that links CD3 on T cells to CD19 on leukemic cells; it is given as a continuous intravenous infusion over 28 days, produces remission in about 45 percent of relapsed adults and is highly effective at clearing MRD. Its principal toxicities are cytokine release syndrome and neurological events. **Inotuzumab ozogamicin** is an anti-CD22 antibody conjugated to the cytotoxin calicheamicin; it achieves remission in about 80 percent of relapsed patients but carries a risk of hepatic veno-occlusive disease, particularly if followed by transplantation. **CAR T-cell therapy** uses the patient's own T cells, genetically engineered to express a chimeric antigen receptor against CD19, expanded ex vivo and reinfused after lymphodepleting chemotherapy. Tisagenlecleucel is approved for patients up to 25 years with relapsed or refractory B-ALL, and brexucabtagene autoleucel and obecabtagene autoleucel for adults; response rates are 70 to 90 percent, with durable remission in roughly half. Cytokine release syndrome and immune effector cell-associated neurotoxicity syndrome are managed with tocilizumab and corticosteroids.

Treatment of T-ALL

T-ALL is treated with the same backbone, with particular benefit from high-dose methotrexate, cyclophosphamide and prolonged asparaginase. Nelarabine, a T-cell-selective purine analogue, improves outcomes when added to front-line therapy in children and is a mainstay of relapse treatment. Trials of the BCL-2 inhibitor venetoclax, the gamma-secretase inhibitors targeting NOTCH1, and CD7-directed CAR T cells are under way.

Adolescents, young adults and older adults

Adolescents and young adults (roughly 15 to 39 years) have better outcomes when treated on paediatric-inspired protocols with intensive asparaginase and prolonged maintenance than on traditional adult regimens. In older adults, who tolerate intensive chemotherapy poorly, lower-intensity regimens combined with blinatumomab, inotuzumab and, for Ph-positive disease, TKIs plus corticosteroids alone, achieve high remission rates with reduced toxicity. The combination of ponatinib and blinatumomab without chemotherapy has produced molecular remission in the large majority of Ph-positive patients in early studies.

Outcomes

Five-year event-free survival now exceeds 85 percent and overall survival 90 percent in childhood ALL. In adults, long-term survival is roughly 40 to 50 percent overall, rising above 60 to 70 percent for younger adults on paediatric-inspired regimens and for Ph-positive patients treated with modern TKI-based approaches. Relapse remains the principal cause of treatment failure, and MRD-guided intensification and immunotherapy are steadily reducing it.

8. Acute Myeloid Leukemia (AML)

Overview and epidemiology

Acute myeloid leukemia is a clonal disorder of myeloid precursors characterised by the accumulation of blasts in the marrow and blood and by impaired production of normal blood cells. It is the commonest acute leukemia in adults, with a median age at diagnosis of about 68 years, and accounts for about 15 to 20 percent of childhood leukemia. AML arises de novo in most cases, but may follow myelodysplastic or myeloproliferative disease (secondary AML) or prior cytotoxic therapy (therapy-related AML); these forms are biologically distinct and respond less well to treatment.

Biology and classification

AML is defined and subclassified by its genetic lesions. The WHO 2022 classification lists AML with defining genetic abnormalities (each diagnosable without a 20 percent blast threshold), AML myelodysplasia-related (defined by specific cytogenetic or mutational findings or a history of MDS), AML defined by differentiation (the successor to the FAB subtypes, used when no defining genetic lesion is found), and secondary myeloid neoplasms. The major genetically defined entities are:

Genetic entity	Frequency in adult AML	Key features
PML::RARA, t(15;17)	About 5–10%	Acute promyelocytic leukemia; coagulopathy; cured with ATRA and arsenic (see Section 9)
RUNX1::RUNX1T1, t(8;21)	About 5–10%	Core-binding factor leukemia; favourable; sensitive to high-dose cytarabine; KIT mutation worsens outlook
CBFB::MYH11, inv(16) or t(16;16)	About 5–8%	Core-binding factor leukemia; often with abnormal eosinophils; favourable
NPM1 mutation	About 30% (50% of normal-karyotype AML)	Cup-shaped nuclei; favourable if FLT3-ITD absent; menin-dependent; MRD monitored by PCR
CEBPA basic leucine zipper (bZIP) in-frame mutation	About 5%	Favourable; germline CEBPA mutation predisposes to familial AML
KMT2A (MLL) rearrangement, 11q23	About 5–10% (more in therapy-related and infant AML)	Monocytic morphology; generally adverse; menin inhibitors active
DEK::NUP214, t(6;9)	About 1%	Adverse; often with FLT3-ITD and basophilia
MECOM (EVI1) rearrangement, inv(3) or t(3;3)	About 1–2%	Adverse; abnormal megakaryocytes; normal or high platelets
BCR::ABL1	Under 1%	Rare; must be distinguished from CML blast crisis; TKI added to chemotherapy
NUP98 rearrangement	Rare in adults; more common in children	Adverse
Myelodysplasia-related (e.g. complex karyotype, –5/del(5q), –7, ASXL1, SRSF2, SF3B1, U2AF1, STAG2, EZH2, BCOR, RUNX1)	About 20–30%	Adverse; often older patients; considered for CPX-351 or hypomethylating agent with venetoclax

The European LeukemiaNet (ELN) 2022 genetic risk classification groups patients into favourable (core-binding factor leukemias, NPM1 mutation without FLT3-ITD, CEBPA bZIP mutation), intermediate

(FLT3-ITD regardless of NPM1, t(9;11), other abnormalities not classified as adverse) and adverse (t(6;9), t(v;11q23), t(9;22), inv(3), -5/del(5q), -7, -17/abn(17p), complex or monosomal karyotype, myelodysplasia-related gene mutations, TP53 mutation). This classification predicts response to intensive chemotherapy and informs the decision about transplantation. TP53 mutation, present in 10 to 15 percent of AML and in most therapy-related and complex-karyotype cases, carries the worst prognosis of any subgroup and remains the greatest unmet need.

Clinical presentation

Patients typically present over a few weeks with the consequences of marrow failure: fatigue, breathlessness, infection and bleeding. Gum hypertrophy, skin infiltration and CNS involvement point to monocytic subtypes; myeloid sarcoma may occur at any site. Leukostasis is a concern when the white count exceeds $100 \times 10^9/L$. Disseminated intravascular coagulation is most severe in APL but can accompany monocytic AML. Some older patients present with a picture indistinguishable from MDS, with cytopenias and few circulating blasts.

Treatment of AML

The first decision in AML is whether the patient is a candidate for intensive induction chemotherapy. This depends on age, performance status, comorbidity (organ function, especially cardiac and hepatic), and the genetics of the disease, since some adverse subtypes respond poorly to intensive therapy irrespective of fitness. Broadly, most patients under 60 to 65 and fit older patients receive intensive therapy; frail or older patients receive lower-intensity therapy, which has become substantially more effective since the introduction of venetoclax.

Intensive induction

The standard backbone, in use for over four decades, is "7+3": cytarabine by continuous infusion for seven days (100 to $200 \text{ mg/m}^2/\text{day}$) with an anthracycline (daunorubicin 60 to 90 mg/m^2 or idarubicin 12 mg/m^2) on days one to three. Complete remission is achieved in 60 to 80 percent of younger adults and 40 to 60 percent of older adults. Targeted agents are added according to genotype: **midostaurin**, a multikinase inhibitor, for FLT3-mutated AML (ITD or TKD), based on the RATIFY trial; **quizartinib**, a more selective FLT3 inhibitor, for FLT3-ITD AML, based on QuANTUM-First; and **gemtuzumab ozogamicin**, an anti-CD33 antibody–calicheamicin conjugate, for favourable and intermediate-risk disease, where it reduces relapse. **CPX-351**, a liposomal formulation of cytarabine and daunorubicin in a fixed 5:1 molar ratio, improves survival compared with 7+3 in older patients with therapy-related or myelodysplasia-related AML. Fludarabine-containing regimens such as FLAG-Ida are used in some centres, particularly in younger patients or for high-risk disease.

Consolidation

Remission is not cure; without further treatment nearly all patients relapse. Consolidation for favourable-risk disease consists of two to four cycles of intermediate- or high-dose cytarabine (1 to 3 g/m^2 twice daily on days 1, 3 and 5), with continuation of midostaurin or quizartinib in FLT3-mutated cases and gemtuzumab where used in induction. For intermediate-risk disease, the choice between chemotherapy consolidation and allogeneic transplantation is guided by MRD status, donor availability and patient preference; for adverse-risk disease, transplantation in first remission is the standard where feasible. Oral azacitidine (CC-486) as maintenance after chemotherapy improves survival in patients aged 55 and over who do not proceed to transplant; FLT3 inhibitors are continued as maintenance in FLT3-mutated AML, and gilteritinib after transplant reduces relapse in patients with detectable MRD.

Lower-intensity therapy

For patients unfit for intensive chemotherapy, the combination of **venetoclax** (an oral BCL-2 inhibitor) with **azacitidine** (a hypomethylating agent given subcutaneously or intravenously for seven days per 28-day cycle) is the standard of care, based on the VIALE-A trial, which showed a composite remission rate of about 66 percent and a median survival of about 15 months compared with 10 months for azacitidine alone. Responses are particularly deep and durable in NPM1-mutated and IDH-mutated AML, and poor in TP53-mutated and FLT3-ITD disease. Venetoclax with low-dose cytarabine is an alternative. Treatment is continuous in 28-day cycles until progression or intolerance, with careful management of neutropenia (venetoclax duration is often shortened from 28 to 14 or 21 days once remission is achieved). Because venetoclax is metabolised by CYP3A4, dose reduction is required with azole antifungals. Older or unfit patients with IDH1 mutation may receive ivosidenib with azacitidine.

Targeted therapies in relapsed or refractory AML

Gilteritinib, a selective FLT3 inhibitor, improved survival compared with salvage chemotherapy in the ADMIRAL trial and is standard for relapsed FLT3-mutated AML. **Ivosidenib** and **olutasidenib** (IDH1 inhibitors) and **enasidenib** (IDH2 inhibitor) induce differentiation of leukemic blasts, producing responses in about 30 to 40 percent of relapsed IDH-mutated patients; differentiation syndrome is a recognised toxicity. **Menin inhibitors** (revumenib, ziftomenib) disrupt the interaction between menin and KMT2A, on which KMT2A-rearranged and NPM1-mutated leukemias depend; revumenib was approved in 2024 for relapsed KMT2A-rearranged acute leukemia and is being combined with venetoclax-based and intensive regimens in trials. Salvage chemotherapy (FLAG-Ida, MEC, high-dose cytarabine) remains the route to second remission and transplant for fit patients without a targetable lesion.

Allogeneic stem cell transplantation in AML

Transplantation offers the best chance of cure for intermediate- and adverse-risk AML in first remission and for any patient in second remission, through the combined effect of high-dose conditioning and the graft-versus-leukemia immune reaction. Donors may be HLA-matched siblings, matched unrelated donors, haploidentical family members (made feasible by post-transplant cyclophosphamide) or umbilical cord blood. Reduced-intensity conditioning has extended eligibility to patients in their seventies. MRD status before transplant strongly predicts relapse, and post-transplant maintenance with FLT3 inhibitors, hypomethylating agents or donor lymphocyte infusions is used to reduce it.

AML in children

Paediatric AML is treated with intensive anthracycline–cytarabine-based chemotherapy, often with etoposide, gemtuzumab and, in FLT3-mutated cases, sorafenib or gilteritinib. Risk stratification uses cytogenetics and MRD; transplantation is reserved for high-risk disease. Survival is about 65 to 75 percent. Children with Down syndrome have a distinctive AML (myeloid leukemia associated with Down syndrome) that is highly curable with reduced-intensity chemotherapy.

Outcomes

Overall five-year survival in AML is about 30 percent, but this masks enormous variation. Younger adults with favourable-risk genetics achieve cure in 60 to 80 percent of cases; older patients with adverse-risk disease treated non-intensively have a median survival of about a year. The past decade has seen more new drug approvals in AML than the previous four combined, and survival in every age group is improving.

9. Acute Promyelocytic Leukemia (APL)

Acute promyelocytic leukemia, formerly FAB M3, deserves separate treatment because it is at once the most dangerous acute leukemia at presentation and the most curable. It accounts for 5 to 10 percent of

adult AML, with a median age of about 40, lower than other AML subtypes, and a higher incidence in Hispanic populations. It is defined by the fusion of the PML gene on chromosome 15 with the retinoic acid receptor alpha gene (RARA) on chromosome 17, produced by t(15;17)(q24;q21). The PML::RARA fusion protein blocks differentiation at the promyelocyte stage and represses the transcription of genes needed for maturation; pharmacological doses of all-trans retinoic acid (ATRA) overcome this block and induce the leukemic cells to mature, while arsenic trioxide degrades the fusion protein and triggers apoptosis.

Presentation

APL presents with pancytopenia and, characteristically, a severe bleeding tendency out of proportion to the platelet count. The abnormal promyelocytes release tissue factor and other procoagulants, and express annexin II, which accelerates fibrinolysis; the result is a consumptive coagulopathy with low fibrinogen, raised D-dimer and prolonged clotting times. Intracranial and pulmonary haemorrhage are the principal causes of early death, which still occurs in 5 to 10 percent of patients, mostly in the first week. The blood film shows promyelocytes with heavy granulation, bilobed or folded nuclei and bundles of Auer rods (faggot cells); a microgranular variant with a higher white count is recognised.

Diagnosis and immediate management

APL is a medical emergency. When the diagnosis is suspected on morphology, ATRA should be started immediately, before genetic confirmation, and aggressive correction of coagulopathy begun. Confirmation by FISH, PCR or immunofluorescence for PML nuclear bodies follows within hours.

Supportive care in the first days is critical: platelets are transfused to keep the count above $30 \times 10^9/L$, cryoprecipitate or fibrinogen concentrate to keep fibrinogen above 1.5 g/L, and fresh frozen plasma to correct prolonged clotting times. Invasive procedures, including lumbar puncture and central line insertion, are avoided until the coagulopathy is controlled. Leukapheresis is contraindicated because it worsens bleeding.

Treatment

For standard-risk APL (white count at diagnosis $10 \times 10^9/L$ or lower), the combination of ATRA and arsenic trioxide without any cytotoxic chemotherapy is the standard of care, based on the APL0406 and AML17 trials, which showed superior event-free survival and less toxicity than ATRA plus chemotherapy. Induction continues until complete remission (typically four to six weeks), followed by four consolidation cycles of arsenic and ATRA over about seven months. Cure rates exceed 95 percent. For high-risk APL (white count above $10 \times 10^9/L$), ATRA and arsenic are combined with an anthracycline (idarubicin) or gemtuzumab ozogamicin during induction to control the count and reduce the risk of differentiation syndrome; outcomes are only slightly inferior. Oral arsenic (realgar–indigo naturalis formula) is used in some countries as an alternative to intravenous arsenic.

Differentiation syndrome

As leukemic promyelocytes mature under ATRA or arsenic, they release cytokines and infiltrate tissues, producing fever, weight gain, oedema, pleural and pericardial effusions, pulmonary infiltrates, hypotension and renal failure. Differentiation syndrome occurs in about 25 percent of patients, usually within the first two weeks, and can be fatal. It is treated promptly with intravenous dexamethasone 10 mg twice daily, and prophylactic steroids are given to patients with a high white count. Other toxicities of ATRA include headache, dry skin and lips, raised lipids and, rarely, pseudotumour cerebri; arsenic causes QT prolongation (requiring electrolyte correction and ECG monitoring), liver enzyme elevation and neuropathy.

Monitoring and relapse

Molecular remission (undetectable PML::RARA by PCR) is expected at the end of consolidation. Molecular monitoring every three months for two years is recommended in high-risk patients. Relapse is uncommon (under 5 percent) and is treated with arsenic-based re-induction, followed by autologous transplantation if a second molecular remission is achieved, or allogeneic transplantation if not.

10. Chronic Lymphocytic Leukemia (CLL)

Overview and epidemiology

Chronic lymphocytic leukemia is a malignancy of mature, functionally incompetent B lymphocytes that accumulate in the blood, marrow, lymph nodes and spleen. It is the commonest leukemia in Western countries, with a median age at diagnosis of about 70 years and a male-to-female ratio of about 2:1. It is markedly less common in East Asian populations, and this difference persists in migrants, pointing to a genetic basis. When the same disease presents predominantly in lymph nodes with fewer than $5 \times 10^9/L$ circulating clonal cells, it is called small lymphocytic lymphoma. Monoclonal B-cell lymphocytosis (MBL), a clonal population below this threshold without lymphadenopathy or cytopenia, is found in about 5 to 10 percent of people over 60 and progresses to CLL requiring treatment at a rate of about 1 to 2 percent per year.

Biology

CLL cells are small, mature-appearing B cells that express CD5 (a T-cell antigen), CD19, CD23 and CD200, with weak surface immunoglobulin and CD20. Their survival depends on signalling through the B-cell receptor, mediated by Bruton's tyrosine kinase (BTK) and phosphoinositide 3-kinase (PI3K), and on over-expression of the anti-apoptotic protein BCL-2, a consequence of the loss of microRNAs miR-15a and miR-16-1 at 13q14. These dependencies are the targets of modern therapy. Two prognostic features are especially important: the mutational status of the immunoglobulin heavy-chain variable region (IGHV), where unmutated IGHV (98 percent or more homology to germline) indicates a more aggressive, less chemosensitive disease; and abnormalities of TP53, either deletion of 17p or mutation, which confer resistance to chemoimmunotherapy and mandate targeted treatment. Other recurrent lesions include del(13q) (favourable), trisomy 12, del(11q) affecting ATM, and mutations of NOTCH1, SF3B1 and BIRC3.

Presentation

Most patients are asymptomatic at diagnosis and identified by a routine blood count showing lymphocytosis. When symptoms occur they include painless lymph node enlargement in the neck, axillae or groin, fatigue, abdominal fullness from splenomegaly, recurrent infections and B symptoms. About 10 to 25 percent develop autoimmune complications, notably warm autoimmune haemolytic anaemia and immune thrombocytopenia, which must be distinguished from marrow failure because they are treated differently. Hypogammaglobulinaemia and impaired T-cell function predispose to bacterial and viral infection, including herpes zoster, and infection is the leading cause of death. Second cancers, particularly skin cancers, are more frequent. Transformation to an aggressive lymphoma, usually diffuse large B-cell lymphoma (Richter transformation), occurs in 2 to 10 percent of patients and presents with rapid nodal growth, B symptoms and a rising LDH.

Staging and when to treat

Rai stage	Binet stage	Definition
0 (low risk)	A	Lymphocytosis only (Binet A: fewer than three lymphoid areas involved, no cytopenia)
I–II (intermediate)	B	Lymphadenopathy and/or hepatosplenomegaly (Binet B: three or more lymphoid areas, no cytopenia)
III–IV (high risk)	C	Anaemia (haemoglobin under 110 g/L, Rai III / under 100 g/L, Binet C) and/or thrombocytopenia (platelets under $100 \times 10^9/L$)

A defining feature of CLL management is that early treatment of asymptomatic disease does not prolong life, a conclusion confirmed even in the era of targeted therapy by the CLL12 trial of ibrutinib. Patients with early stage disease are therefore followed with clinical review and blood counts every three to twelve months, an approach called watch and wait or active monitoring. Treatment is started when the International Workshop on CLL (iwCLL) criteria for active disease are met: progressive marrow failure, massive or progressive splenomegaly or lymphadenopathy, lymphocyte doubling time under six months, autoimmune cytopenia not responding to steroids, or disabling constitutional symptoms. A rising lymphocyte count alone is not an indication.

Treatment of CLL

Treatment of CLL has changed more in the past decade than that of any other leukemia. Chemoimmunotherapy has been almost entirely replaced by two classes of oral targeted agent, used alone or in combination, and choice between them depends on genetics, comorbidity and patient preference regarding continuous versus fixed-duration therapy.

BTK inhibitors

Ibrutinib, the first-in-class covalent BTK inhibitor, produced durable remissions in patients including those with del(17p) and unmutated IGHV and superseded chemoimmunotherapy in front-line trials. Its off-target effects (atrial fibrillation, hypertension, bleeding, arthralgia) led to the development of second-generation covalent inhibitors, acalabrutinib and zanubrutinib, which have equivalent efficacy with fewer cardiovascular toxicities and are now preferred. BTK inhibitors are taken continuously, produce a transient rise in lymphocyte count as cells leave the nodes (redistribution lymphocytosis, which is not progression), and achieve durable disease control in most patients for many years, though complete remissions are uncommon and MRD usually remains detectable. Resistance arises through mutations of BTK (C481S) or PLCG2. Pirtobrutinib, a non-covalent BTK inhibitor, is active against C481-mutated disease and is approved for patients who have progressed on covalent BTK inhibitors and venetoclax.

BCL-2 inhibition: venetoclax-based fixed-duration therapy

Venetoclax induces rapid apoptosis of CLL cells and, combined with an anti-CD20 antibody, produces deep remissions with undetectable MRD in the majority of patients. The CLL14 trial established one year of venetoclax plus obinutuzumab as a front-line fixed-duration option, with progression-free survival exceeding six years in most patients; the MURANO trial established two years of venetoclax plus rituximab for relapsed disease. Because tumour lysis is a real risk, venetoclax is started with a five-week dose ramp-up from 20 mg to 400 mg daily, with hydration, uric acid prophylaxis and, for patients with bulky nodes or high counts, in-hospital monitoring. Fixed-duration therapy is attractive because it avoids the cumulative toxicity, cost and resistance of indefinite treatment, and can be repeated at relapse.

Combined targeted therapy

All-oral, fixed-duration combinations of a BTK inhibitor with venetoclax, such as ibrutinib–venetoclax (GLOW and CAPTIVATE trials) and acalabrutinib–venetoclax with or without obinutuzumab (AMPLIFY trial), achieve very high rates of undetectable MRD and are approved or emerging as front-line options, including MRD-guided treatment duration.

Chemoimmunotherapy

Fludarabine, cyclophosphamide and rituximab (FCR) can produce very long remissions, possibly amounting to cure, in younger fit patients with mutated IGHV and no TP53 abnormality, and remains an option in that narrow group where targeted agents are unavailable. Bendamustine–rituximab and chlorambucil–obinutuzumab were used in older patients but have been displaced by targeted therapy, which is superior in every comparison. Chemoimmunotherapy is contraindicated in TP53-disrupted disease.

Choosing front-line therapy

Patient group	Preferred approach
TP53 deleted or mutated	Continuous BTK inhibitor (acalabrutinib or zanubrutinib) or venetoclax–obinutuzumab; BTK inhibitor generally preferred for durability; avoid chemoimmunotherapy
Unmutated IGHV, no TP53 abnormality	BTK inhibitor or venetoclax-based fixed-duration therapy; BTK inhibitor–venetoclax combination increasingly used
Mutated IGHV, no TP53 abnormality	Venetoclax–obinutuzumab fixed duration (deep, durable remission); BTK inhibitor; FCR in selected young fit patients
Significant cardiac disease or anticoagulation	Venetoclax-based therapy (avoids BTK inhibitor cardiovascular and bleeding risk)
Impaired renal function or high tumour lysis risk	BTK inhibitor (avoids venetoclax ramp-up risk)

Relapsed disease and cellular therapy

Patients who relapse after fixed-duration venetoclax can be retreated with venetoclax or switched to a BTK inhibitor, and vice versa. Double-refractory disease (progression on both classes) is treated with pirtobrutinib, the PI3K inhibitors idelalisib or duvelisib (limited by immune-mediated colitis, hepatitis and pneumonitis), or CD19 CAR T-cell therapy, for which lisocabtagene maraleucel was approved in 2024. Allogeneic transplantation is reserved for fit patients with high-risk, multiply relapsed disease. Richter transformation is treated as an aggressive lymphoma, usually with R-CHOP-type chemotherapy followed by transplant, though outcomes remain poor and clinical trials of bispecific antibodies and CAR T cells are encouraged.

Supportive measures specific to CLL

Vaccination against pneumococcus, influenza, COVID-19 and recombinant zoster (never live vaccines) is recommended at diagnosis, before treatment blunts the response. Immunoglobulin replacement is given to patients with hypogammaglobulinaemia and recurrent serious infections. Regular skin surveillance is advised. Autoimmune cytopenias respond to corticosteroids, rituximab or, when they occur on treatment, to treating the CLL itself.

Outcomes

Median survival from diagnosis exceeds ten years overall and is substantially longer for early-stage and favourable-genetics disease; many patients with CLL die of unrelated causes. For those who need treatment, targeted therapy has extended survival to the point where, for the majority, CLL is a chronic

condition managed over decades.

11. Chronic Myeloid Leukemia (CML)

Overview and biology

Chronic myeloid leukemia is a myeloproliferative neoplasm in which a pluripotent stem cell acquires the Philadelphia chromosome, a shortened chromosome 22 produced by the reciprocal translocation t(9;22)(q34;q11). This fuses the BCR gene on chromosome 22 with the ABL1 gene on chromosome 9. The BCR::ABL1 fusion protein (usually p210, occasionally p190 or p230) is a constitutively active tyrosine kinase that drives proliferation, inhibits apoptosis and impairs adhesion of myeloid progenitors. CML was the first cancer linked to a specific chromosomal abnormality (1960) and the first to be treated with a rationally designed molecularly targeted drug (imatinib, 2001), an achievement that transformed a disease with a median survival of about five years into one in which most patients have a normal life expectancy. It accounts for about 15 percent of adult leukemia, with a median age at diagnosis of 55 to 65 years, though it occurs at all ages including childhood.

Phases

Untreated CML progresses through three phases. About 90 percent of patients are diagnosed in **chronic phase**, characterised by leucocytosis with a full spectrum of maturing granulocytes, basophilia, often thrombocytosis, splenomegaly and fewer than 10 percent blasts. **Accelerated phase** is marked by 10 to 19 percent blasts, basophils of 20 percent or more, persistent thrombocytopenia or thrombocytosis unresponsive to therapy, increasing spleen size, or new cytogenetic abnormalities (clonal evolution); the WHO 2022 classification has de-emphasised this phase in favour of identifying high-risk features. **Blast phase**, with 20 percent or more blasts in blood or marrow or extramedullary blast proliferation, resembles acute leukemia (myeloid in about two thirds of cases, lymphoid in one third) and carries a poor prognosis. Progression to blast phase, once nearly universal, is now rare in patients who respond to tyrosine kinase inhibitors.

Presentation and diagnosis

Up to half of patients are asymptomatic at diagnosis. Others present with fatigue, weight loss, night sweats, left upper quadrant discomfort from splenomegaly, gout, or symptoms of hyperviscosity. The blood film is highly characteristic; the diagnosis is confirmed by demonstration of BCR::ABL1 by PCR or FISH on blood, and a marrow examination with karyotype is performed to assess phase and detect additional abnormalities. Baseline risk is scored using the Sokal, Hasford (Euro) or, in the TKI era, ELTS scores, which combine age, spleen size, platelet count and blast and basophil percentages.

Treatment of CML

Tyrosine kinase inhibitors

Oral TKIs that bind the ATP pocket of BCR::ABL1 are the standard first-line treatment in all phases. Five are approved for chronic phase: imatinib (first generation), dasatinib, nilotinib and bosutinib (second generation), and asciminib, a first-in-class allosteric inhibitor that binds the myristoyl pocket (STAMP inhibitor) and was approved for newly diagnosed disease in 2024 after the ASC4FIRST trial showed superior molecular response with better tolerability. Ponatinib, a third-generation inhibitor active against the T315I gatekeeper mutation, is reserved for resistant disease because of arterial occlusive toxicity.

Agent	Usual dose	Characteristic considerations
Imatinib	400 mg once daily	Longest safety record; fluid retention, muscle cramps, gastrointestinal upset, rash; slower molecular response than second-generation agents but equivalent long-term survival
Dasatinib	100 mg once daily	Faster responses; pleural effusion (in up to 30% over time), pulmonary arterial hypertension (rare), bleeding
Nilotinib	300 mg twice daily (fasting)	Faster responses; arterial occlusive events, QT prolongation, hyperglycaemia, pancreatitis, hepatic enzyme rise
Bosutinib	400 mg once daily	Diarrhoea (usually early and self-limiting), hepatotoxicity
Asciminib	80 mg once daily or 40 mg twice daily; 200 mg twice daily for T315I	Better tolerated; lipase elevation, hypertension; active against T315I at higher dose
Ponatinib	45 mg reduced to 15 mg on response	Only agent besides high-dose asciminib active against T315I; arterial and venous thrombosis, hypertension, hepatotoxicity

Monitoring response

Response is measured by quantitative PCR for BCR::ABL1 on peripheral blood every three months, expressed on the International Scale. The European LeukemiaNet 2020 milestones define optimal response as BCR::ABL1 of 10 percent or less at three months, 1 percent or less at six months, and 0.1 percent or less (major molecular response, MMR) at twelve months; failure to reach these milestones warrants mutation analysis and a change of TKI. Deeper responses are designated MR4 (0.01 percent), MR4.5 (0.0032 percent) and MR5. Adherence is the commonest cause of poor response and should always be explored before attributing failure to resistance.

Resistance and switching

Resistance may result from point mutations in the kinase domain (T315I confers resistance to all first- and second-generation agents), BCR::ABL1 amplification, or BCR::ABL1-independent mechanisms. Intolerance is a commoner reason for switching. Choice of second-line agent is guided by the mutation profile and the patient's comorbidities: for example, avoiding nilotinib and ponatinib in patients with vascular disease and dasatinib in those with lung disease. Allogeneic transplantation, once the only curative therapy, is now reserved for patients who fail multiple TKIs, have T315I disease not controlled by ponatinib or asciminib, or present in or progress to accelerated or blast phase.

Treatment-free remission

Patients who have sustained a deep molecular response (at least MR4) for a minimum of two years, after at least three to five years of TKI therapy, may attempt to stop treatment under close molecular monitoring (monthly PCR for the first six months, then every two to three months). About half maintain remission without treatment; the remainder lose MMR, usually within six months, and regain it on resuming the TKI. Treatment-free remission is now a formal goal of therapy, and is a particular consideration for younger patients, women planning pregnancy (TKIs are teratogenic) and those with troublesome side effects. A transient withdrawal syndrome of musculoskeletal pain occurs in about 20 to 30 percent.

Blast phase

Blast phase is treated as acute leukemia with the addition of a TKI (usually ponatinib or a second-generation agent chosen by mutation status): AML-type induction for myeloid blast phase, ALL-type induction for lymphoid blast phase, followed as soon as possible by allogeneic transplantation, which is the only treatment that offers a realistic prospect of long-term survival. Blinatumomab has

activity in lymphoid blast phase.

Outcomes

With TKI therapy, ten-year overall survival in chronic-phase CML exceeds 85 to 90 percent, and life expectancy approaches that of the general population. The main challenges are now the cost and cumulative toxicity of lifelong therapy, cardiovascular risk management, and expanding access to treatment-free remission.

12. Less Common Leukemias

Hairy cell leukemia

Hairy cell leukemia is an indolent B-cell malignancy of middle-aged and older adults, four times commoner in men, accounting for about 2 percent of leukemias. The cells have characteristic hair-like cytoplasmic projections, express CD11c, CD25, CD103 and CD123, and in almost all classical cases carry the BRAF V600E mutation. Patients present with pancytopenia, particularly monocytopenia, massive splenomegaly and recurrent infections, including atypical mycobacteria. Marrow aspiration is frequently dry because of reticulin fibrosis, and the trephine shows a characteristic fried-egg appearance. Treatment is indicated for symptomatic cytopenias or splenomegaly and consists of a single course of a purine analogue, cladribine (5 to 7 days) or pentostatin, which produces complete remission in 80 to 90 percent of patients lasting many years; the addition of rituximab deepens and prolongs response. Relapsed disease is retreated with a purine analogue and rituximab, or with the BRAF inhibitor vemurafenib (with or without rituximab), the anti-CD22 immunotoxin moxetumomab pasudotox, or BTK inhibitors. The variant form (hairy cell leukemia variant, now called splenic B-cell lymphoma/leukemia with prominent nucleoli) lacks BRAF mutation and CD25, presents with a high white count, and responds poorly to cladribine alone.

Chronic myelomonocytic leukemia (CMML)

CMML is a myelodysplastic/myeloproliferative neoplasm defined by a persistent absolute monocytosis of at least $0.5 \times 10^9/L$ constituting 10 percent or more of white cells, dysplasia, fewer than 20 percent blasts and absence of BCR::ABL1. It affects older adults (median age 70 to 75), and presents either with a proliferative picture (high white count, splenomegaly, B symptoms) or a dysplastic picture (cytopenias). Mutations of TET2, SRSF2, ASXL1 and RAS-pathway genes are characteristic. About 15 to 30 percent transform to AML. Treatment is risk-adapted: hydroxyurea for proliferative symptoms, hypomethylating agents (azacitidine, decitabine) for higher-risk or dysplastic disease, and allogeneic transplantation for fit patients with high-risk features, which is the only curative option.

Juvenile myelomonocytic leukemia (JMML)

JMML is a rare, aggressive myeloproliferative disorder of infancy and early childhood driven by hyperactivation of the RAS pathway through mutations of PTPN11, NRAS, KRAS, CBL or NF1. It occurs at increased frequency in children with neurofibromatosis type 1 and Noonan syndrome. Children present with fever, pallor, rash, hepatosplenomegaly, monocytosis and raised fetal haemoglobin. Chemotherapy is ineffective; allogeneic transplantation cures about 50 to 60 percent. Some CBL-mutated and Noonan-associated cases resolve spontaneously. MEK inhibitors are under investigation.

Prolymphocytic leukemias

T-cell prolymphocytic leukemia (T-PLL) is a rare, aggressive mature T-cell leukemia of older adults, with very high lymphocyte counts, hepatosplenomegaly, lymphadenopathy, skin lesions and effusions. Inversion of chromosome 14 involving TCL1 and mutations of ATM and JAK-STAT genes are typical. The anti-CD52 antibody alemtuzumab, given intravenously, produces remission in most patients, but relapse is near-universal without consolidation by allogeneic transplantation. **B-cell prolymphocytic leukemia** is now largely reclassified as other entities in WHO 2022 (mantle cell lymphoma, splenic lymphoma) and is treated with regimens appropriate to those diseases, including BTK inhibitors.

Large granular lymphocytic leukemia (LGL)

LGL leukemia is an indolent clonal proliferation of cytotoxic T cells (85 percent) or NK cells, frequently associated with autoimmune disease (especially rheumatoid arthritis) and driven by STAT3 or STAT5B mutations. It presents with neutropenia, anaemia, recurrent infection and splenomegaly. Treatment is indicated only for symptomatic cytopenia and consists of immunosuppression with low-dose methotrexate, cyclophosphamide or ciclosporin; responses are slow, and many patients need no therapy. Aggressive NK-cell leukemia, by contrast, is a fulminant EBV-associated disease of young adults in East Asia and requires intensive treatment and transplantation.

Adult T-cell leukemia/lymphoma (ATLL)

ATLL is caused by HTLV-1 infection and presents in acute, lymphomatous, chronic or smouldering forms. The acute form features a high count of flower-shaped lymphocytes, hypercalcaemia, lytic bone lesions, skin lesions and opportunistic infection, and has a median survival of under a year. Treatment of aggressive forms combines intensive chemotherapy with the anti-CCR4 antibody mogamulizumab, or, in some countries, interferon-alpha with zidovudine; allogeneic transplantation offers the only prospect of cure. Indolent forms may be observed.

Mixed phenotype acute leukemia (MPAL)

MPAL comprises about 2 to 3 percent of acute leukemias and shows blasts expressing definitive markers of two lineages, most commonly B/myeloid or T/myeloid. Some cases carry BCR::ABL1 or KMT2A rearrangement. It is treated with ALL-type induction (which is more effective than AML-type), a TKI where BCR::ABL1 is present, and allogeneic transplantation in first remission for most adults.

Blastic plasmacytoid dendritic cell neoplasm (BPDCN)

BPDCN is a rare aggressive malignancy, often presenting with skin lesions and marrow involvement, expressing CD4, CD56 and CD123. The CD123-directed immunotoxin tagraxofusp produces high response rates and is used as a bridge to allogeneic transplantation; ALL-type chemotherapy and venetoclax-based regimens are alternatives.

Leukemia in special situations

Infant ALL (under one year) is usually KMT2A-rearranged, presents with very high white counts and CNS involvement, and has been the most difficult childhood leukemia to cure; the addition of blinatumomab to chemotherapy has dramatically improved outcomes in recent trials. **Leukemia in pregnancy** requires multidisciplinary care; chemotherapy is avoided in the first trimester where possible, and anthracycline-based regimens are relatively safe thereafter, while TKIs, arsenic and methotrexate are teratogenic. **Myeloid sarcoma** is treated as AML with systemic chemotherapy even when disease appears localised.

13. Principles of Treatment

Although each leukemia has its own regimen, a common set of principles underlies all of them. Understanding these principles helps make sense of why treatment is structured as it is.

Curative versus disease-controlling intent

Acute leukemias are treated with curative intent whenever the patient can tolerate the therapy required. Chronic leukemias are mostly managed with disease control as the goal, accepting that the disease may persist at low levels for decades, although treatment-free remission in CML and MRD-negative fixed-duration therapy in CLL have blurred this distinction. The intent of treatment should be explicit at the outset, revisited as the disease evolves and shared openly with the patient.

Phased therapy

Curative treatment of acute leukemia is divided into induction (achieving remission), consolidation (eradicating residual disease with different agents to circumvent resistance) and, in ALL, maintenance (prolonged low-dose therapy to suppress re-emergence of the clone). The rationale is the fractional cell kill hypothesis: each course kills a fixed proportion rather than a fixed number of cells, so repeated courses are needed to reduce a population of 10^{12} cells at diagnosis to zero.

Risk-adapted and response-adapted therapy

Treatment intensity is matched to the risk of relapse, so that patients with favourable disease are spared unnecessary toxicity and those with adverse disease receive the intensification, including transplantation, that they need. Risk is assessed at diagnosis from clinical and genetic features and then refined by the response to treatment, of which MRD is the most sensitive measure. This is the single most important advance in leukemia therapy of the past thirty years and explains why childhood ALL survival rose from under 10 percent in the 1960s to over 90 percent today largely with the same drugs.

Modes of action of the principal drug classes

Class	Examples	Mechanism
Antimetabolites	Cytarabine, methotrexate, mercaptopurine, thioguanine, fludarabine, cladribine, nelarabine, clofarabine	Mimic nucleotides and disrupt DNA synthesis in dividing cells
Anthracyclines	Daunorubicin, idarubicin, doxorubicin, mitoxantrone	Intercalate DNA and inhibit topoisomerase II; cumulative cardiotoxicity
Alkylating agents	Cyclophosphamide, busulfan, melphalan, bendamustine, chlorambucil	Cross-link DNA; used in ALL, CLL and transplant conditioning
Vinca alkaloids	Vincristine	Disrupt microtubules and mitosis; peripheral neuropathy
Asparaginase	Pegaspargase, Erwinia asparaginase	Deprive lymphoblasts of asparagine, which they cannot synthesise
Glucocorticoids	Prednisolone, dexamethasone	Induce apoptosis in lymphoid cells
Hypomethylating agents	Azacitidine, decitabine	Inhibit DNA methyltransferase; restore expression of silenced genes
Differentiating agents	ATRA, arsenic trioxide, IDH inhibitors	Release the maturation block in APL and IDH-mutated AML

Class	Examples	Mechanism
Kinase inhibitors	Imatinib, dasatinib, nilotinib, bosutinib, ponatinib, asciminib (BCR::ABL1); midostaurin, gilteritinib, quizartinib (FLT3); ibrutinib, acalabrutinib, zanubrutinib, pirtobrutinib (BTK)	Block the specific enzyme on which the leukemic clone depends
BCL-2 inhibitor	Venetoclax	Restores apoptosis by neutralising the survival protein BCL-2
Menin inhibitors	Revumenib, ziftomenib	Disrupt menin–KMT2A interaction in KMT2A-rearranged and NPM1-mutated leukemia
Monoclonal antibodies	Rituximab, obinutuzumab (CD20); alemtuzumab (CD52); mogamulizumab (CCR4)	Direct cytotoxicity, complement and immune cell-mediated killing
Antibody–drug conjugates	Inotuzumab ozogamicin (CD22), gemtuzumab ozogamicin (CD33)	Deliver a cytotoxin selectively to antigen-bearing cells
Bispecific T-cell engagers	Blinatumomab (CD19×CD3)	Physically link T cells to leukemic cells
CAR T cells	Tisagenlecleucel, brexucabtagene, obecabtagene, lisocabtagene (CD19)	Patient's own T cells engineered to recognise and kill leukemic cells

Haematopoietic stem cell transplantation

Transplantation replaces the patient's haematopoietic system after high-dose conditioning. In **autologous** transplantation the patient's own previously harvested stem cells are reinfused; it permits dose intensification but provides no immune effect against leukemia and is used mainly in relapsed APL. In **allogeneic** transplantation, stem cells from a donor are infused, and the donor immune system mounts a graft-versus-leukemia reaction that is the principal mechanism of cure; the same immune reaction may attack the patient's tissues, causing graft-versus-host disease (GVHD). Conditioning may be myeloablative (high-dose busulfan or cyclophosphamide, or total body irradiation) or reduced-intensity (fludarabine with lower-dose alkylator), the latter relying more on the immune effect and tolerable in older patients. Donor sources include matched siblings (about 25 to 30 percent of patients have one), matched unrelated volunteers from international registries, haploidentical relatives (parents, children, half-matched siblings), and cord blood. Post-transplant cyclophosphamide has made haploidentical transplantation safe and widely available.

Complications include early infection during neutropenia, mucositis, sinusoidal obstruction syndrome, acute GVHD (skin, gut, liver) in the first 100 days, chronic GVHD thereafter, CMV reactivation, and late effects such as infertility, endocrine dysfunction and secondary cancers. Transplant-related mortality has fallen to about 10 to 20 percent with modern practice. Relapse after transplant is managed with withdrawal of immunosuppression, donor lymphocyte infusion, targeted agents or second transplantation.

Clinical trials

Almost every advance in leukemia treatment has come from a cooperative clinical trial, and enrolment in a trial is recommended as the preferred option whenever one is available. Trials are particularly important in relapsed disease, TP53-mutated AML, older adults with ALL, and Richter transformation, where standard treatment is unsatisfactory.

14. Treatment by Type: A Consolidated View

Leukemia	First-line treatment	Relapsed or refractory	Role of transplant
Childhood ALL	Multi-agent chemotherapy over 2–3 years (induction, consolidation, delayed intensification, maintenance) with intrathecal CNS prophylaxis; TKI if Ph-positive; blinatumomab increasingly incorporated	Chemotherapy re-induction; blinatumomab; inotuzumab; CD19 CAR T cells (tisagenlecleucel)	High-risk first remission (persistent MRD, adverse genetics, induction failure); second remission
Adult ALL	Paediatric-inspired chemotherapy (younger adults) or reduced-intensity plus immunotherapy (older adults); blinatumomab consolidation; rituximab if CD20-positive; TKI (ponatinib/dasatinib) if Ph-positive	Blinatumomab, inotuzumab, CAR T cells (brexucabtagene, obecabtagene), nelarabine for T-ALL, salvage chemotherapy	High-risk first remission; most patients in second remission
AML (fit)	7+3 ± midostaurin/quizartinib (FLT3), ± gemtuzumab (favourable/intermediate); CPX-351 for therapy-related or MDS-related; high-dose cytarabine consolidation	Gilteritinib (FLT3), IDH inhibitors, menin inhibitors (KMT2A, NPM1), salvage chemotherapy, venetoclax combinations	Intermediate- and adverse-risk first remission; all second remissions
AML (unfit)	Azacitidine plus venetoclax (continuous cycles); ivosidenib plus azacitidine if IDH1-mutated; low-dose cytarabine plus venetoclax	Targeted agents by mutation; clinical trial; best supportive care	Reduced-intensity transplant in selected responding patients
APL	ATRA plus arsenic trioxide (standard risk); add idarubicin or gemtuzumab for high risk; aggressive coagulopathy support	Arsenic-based re-induction	Autologous in second molecular remission; allogeneic if not achieved
CLL	Watch and wait until iwCLL criteria met; then BTK inhibitor (acalabrutinib, zanubrutinib) continuous, or venetoclax–obinutuzumab fixed duration, or BTK inhibitor–venetoclax combination; targeted therapy mandatory if TP53-disrupted	Switch class; pirtobrutinib; venetoclax retreatment; PI3K inhibitors; CD19 CAR T cells (lisocabtagene)	Reserved for fit patients with multiply relapsed high-risk disease or Richter transformation
CML (chronic phase)	TKI: imatinib, dasatinib, nilotinib, bosutinib or asciminib, chosen by comorbidity; lifelong or until treatment-free remission criteria met	Switch TKI by mutation and tolerance; ponatinib or high-dose asciminib for T315I	After failure of multiple TKIs or progression to advanced phase
CML (blast phase)	Acute leukemia-type induction plus TKI (ponatinib or by mutation)	Clinical trial	Standard for all eligible patients as soon as remission achieved
Hairy cell leukemia	Cladribine or pentostatin ± rituximab when symptomatic	Repeat purine analogue plus rituximab; vemurafenib; moxetumomab; BTK inhibitor	Rarely required
CMML	Hydroxyurea (proliferative), hypomethylating agents (higher risk)	Clinical trial	Only curative option for fit high-risk patients
JMML	Allogeneic transplant (chemotherapy ineffective); observation for some CBL/Noonan cases	Second transplant; MEK inhibitors in trials	Primary treatment
T-PLL	Alemtuzumab	Alemtuzumab retreatment; trials	Consolidation in first remission for all eligible patients

15. Supportive Care and Managing Complications

Much of the improvement in leukemia survival over the past fifty years is attributable not to new anti-leukemic drugs but to better supportive care: the ability to keep patients alive and well through weeks of profound pancytopenia and through the toxicities of treatment. Supportive care is not an adjunct to leukemia therapy; it is an integral part of it.

Infection prevention and management

Neutropenia (absolute neutrophil count under $0.5 \times 10^9/\text{L}$) following intensive chemotherapy lasts two to four weeks and is the period of greatest danger. Prophylaxis includes a fluoroquinolone (levofloxacin) against bacterial infection, an azole antifungal (posaconazole is preferred in AML induction because of its activity against moulds), aciclovir or valaciclovir against herpes simplex and zoster, and cotrimoxazole against *Pneumocystis* in ALL and in patients on prolonged steroids or purine analogues. Hepatitis B carriers require antiviral prophylaxis before rituximab or intensive therapy.

Neutropenic fever (a single temperature of 38.3°C or a sustained 38.0°C in a neutropenic patient) is treated as sepsis until proved otherwise: blood cultures from peripheral vein and central line, then broad-spectrum intravenous antibiotics (typically piperacillin–tazobactam or meropenem) within one hour, without waiting for results. Persistent fever beyond four to seven days prompts empirical or CT-guided antifungal therapy for invasive aspergillosis. Granulocyte colony-stimulating factor (G-CSF) shortens neutropenia after ALL chemotherapy and in consolidation for AML, but is generally withheld during AML induction where it may stimulate leukemic blasts.

Transfusion support

Red cells are transfused to maintain haemoglobin above about 70 to 80 g/L, higher in patients with cardiac disease or symptoms. Platelets are given prophylactically when the count falls below $10 \times 10^9/\text{L}$, or below $20 \times 10^9/\text{L}$ with fever or infection, or higher thresholds before procedures and in APL. All products should be leucodepleted; irradiated products are required for patients receiving purine analogues, alemtuzumab, or undergoing transplantation, to prevent transfusion-associated GVHD. CMV-negative or leucodepleted products are used for transplant candidates. Iron overload from repeated transfusion is monitored by ferritin.

Tumour lysis syndrome

Patients with high white counts, bulky disease, high LDH or renal impairment, and all patients beginning venetoclax, are at risk. Prevention consists of aggressive intravenous hydration, allopurinol (which blocks uric acid production) or, in high-risk cases, rasburicase (recombinant urate oxidase, which degrades existing uric acid within hours), and monitoring of potassium, phosphate, calcium, uric acid and creatinine every six to twelve hours during the first days of treatment. Established tumour lysis may require renal replacement therapy.

Hyperleucocytosis and leukostasis

Cytoreduction with hydroxyurea and prompt initiation of definitive therapy is the primary approach. Leukapheresis can rapidly lower the count in symptomatic leukostasis but has not been shown to improve survival, is contraindicated in APL, and should not delay chemotherapy. Red cell transfusion is minimised until the count falls, as it increases viscosity.

Bleeding and coagulopathy

Beyond platelet transfusion, disseminated intravascular coagulation is managed with fibrinogen replacement and plasma as described for APL. Tranexamic acid is used cautiously for mucosal bleeding but avoided in DIC. Menstrual suppression with hormonal therapy is offered to women undergoing intensive treatment.

Nausea, mucositis and nutrition

Modern antiemetic regimens (5-HT₃ antagonists, NK1 antagonists, dexamethasone, olanzapine) control chemotherapy-induced nausea in most patients. Oral mucositis from high-dose cytarabine, methotrexate and conditioning is managed with meticulous oral hygiene, cryotherapy, topical and systemic analgesia, and attention to infection. Dietitian input, and where necessary enteral or parenteral nutrition, maintain weight during prolonged admission.

Drug-specific toxicities

- **Anthracyclines:** cumulative cardiomyopathy (lifetime daunorubicin above about 550 mg/m²); baseline and periodic echocardiography; dexrazoxane in children.
- **Asparaginase:** pancreatitis, thrombosis (including cerebral venous sinus thrombosis), hyperglycaemia, hepatotoxicity, hypersensitivity; antithrombin monitoring and switch to Erwinia preparation on allergy.
- **Vincristine:** peripheral neuropathy, constipation, jaw pain; dose capping; never intrathecal (fatal).
- **High-dose methotrexate:** nephrotoxicity and mucositis; urinary alkalinisation, leucovorin rescue, level monitoring; glucarpidase for toxic levels.
- **High-dose cytarabine:** cerebellar toxicity (assess before each dose), conjunctivitis (steroid eye drops).
- **Corticosteroids:** hyperglycaemia, myopathy, mood change, avascular necrosis of bone (particularly in adolescents), osteoporosis, infection.
- **Blinatumomab and CAR T cells:** cytokine release syndrome (fever, hypotension, hypoxia; tocilizumab) and neurotoxicity (confusion, aphasia, seizures; corticosteroids); pre-treatment cyto-reduction and step-up dosing reduce risk.
- **Inotuzumab and gemtuzumab:** hepatic sinusoidal obstruction syndrome, particularly around transplantation; defibrotide for treatment.
- **Venetoclax:** tumour lysis, neutropenia; CYP3A4 interactions with azoles.
- **BTK inhibitors:** atrial fibrillation, hypertension, bleeding (hold around surgery), arthralgia; fewer with second-generation agents.
- **TKIs for CML:** see Section 11; drug interactions via CYP3A4 are common.
- **Hypomethylating agents:** cytopenias, injection-site reactions, nausea.

Central venous access

Most patients undergoing intensive therapy have a tunnelled central venous catheter or peripherally inserted central catheter for chemotherapy, blood products, antibiotics and sampling. Line infection and thrombosis are the main complications.

Psychological, social and palliative support

A diagnosis of leukemia is life-altering. Patients and families benefit from early involvement of clinical nurse specialists, psychologists, social workers and, for children, play specialists and educational support. Fatigue, anxiety, depression and financial strain are common and should be asked about at

every visit. Palliative care expertise is valuable throughout, not only at the end of life, for symptom control and decision-making. Where disease is refractory, honest discussion of prognosis, priorities and preferences, including the option of supportive care alone, is part of good treatment.

16. Prognosis, Follow-up and Survivorship

Determinants of prognosis

Across all leukemias, prognosis is shaped by three sets of factors: the biology of the disease (genetics, subtype, secondary versus de novo), the patient (age, fitness, comorbidity, and social circumstances that affect adherence and access), and the response to treatment (speed and depth of remission, MRD status). Of these, genetics and MRD response have become the dominant considerations in acute leukemia.

Leukemia	Approximate long-term survival with current therapy
Childhood ALL	Over 90% five-year survival; over 85% event-free
Adult ALL	About 40–50% overall; 60–70% in young adults on paediatric-inspired regimens; improving with immunotherapy
AML, younger adults, favourable risk	60–80% cured
AML, younger adults, intermediate/adverse risk	30–50% / 15–30% cured, depending on transplant
AML, older adults treated non-intensively	Median survival about 12–18 months; long-term survival uncommon but rising
APL	Over 90–95% cured; early death the main threat
CLL	Median survival exceeds 10–15 years; many patients never require treatment
CML, chronic phase	Near-normal life expectancy with TKI therapy
Hairy cell leukemia	Near-normal life expectancy

Follow-up after treatment

After completion of therapy for acute leukemia, patients are reviewed with blood counts monthly for the first year, then at increasing intervals to five years; marrow examination and MRD testing are performed where clinically indicated or according to protocol (for example NPM1 or PML::RARA monitoring). The risk of relapse is highest in the first two years and becomes very low after five years. Patients on continuous targeted therapy for CLL or CML are reviewed every three to six months with attention to adherence, toxicity, cardiovascular risk and, in CML, molecular response.

Late effects

Survivors of childhood ALL, who now number in the hundreds of thousands, may experience long-term consequences: neurocognitive effects of intrathecal and high-dose methotrexate (and of cranial irradiation in older cohorts), obesity and metabolic syndrome, reduced bone density, avascular necrosis, cardiomyopathy from anthracyclines, peripheral neuropathy and, after transplantation, infertility, endocrine deficiencies, cataract and chronic GVHD. Secondary cancers, including therapy-related myeloid neoplasms and solid tumours, occur in a small percentage of survivors. Structured long-term follow-up programmes, with screening tailored to the treatments received, are recommended, together with attention to healthy lifestyle, vaccination (revaccination is required after transplantation) and

psychosocial adjustment, including return to education or work.

Living with chronic leukemia

For patients with CLL and CML, the challenge is less survival than long-term quality of life: managing the side effects of continuous oral therapy, the financial and psychological burden of a lifelong diagnosis, the increased susceptibility to infection and second cancers, and uncertainty about the future. Clear information, shared decision-making about when to start, switch or stop treatment, and access to peer support all help.

17. Glossary of Terms

Term	Meaning
Allogeneic transplant	Transplantation of stem cells from another person (donor).
Anaemia	A shortage of red blood cells or haemoglobin, causing fatigue and breathlessness.
Auer rod	A needle-shaped inclusion in the cytoplasm of myeloid blasts; diagnostic of myeloid lineage.
Autologous transplant	Transplantation of the patient's own previously collected stem cells.
B symptoms	Fever, drenching night sweats and unintentional weight loss.
BCR::ABL1	The fusion gene created by the Philadelphia chromosome; the driver of CML and some ALL.
Blast	An immature blood cell; the malignant cell of acute leukemia.
Bone marrow	The soft tissue inside bones where blood cells are produced.
CAR T cells	Chimeric antigen receptor T cells; a patient's T cells engineered to attack leukemia.
Chemoimmunotherapy	Chemotherapy combined with a monoclonal antibody such as rituximab.
Clonal	Derived from a single ancestral cell; a defining feature of cancer.
Complete remission	Fewer than 5% blasts in the marrow with recovery of normal blood counts and no extramedullary disease.
Consolidation	Treatment given after remission to eliminate residual disease.
Cytogenetics	Study of chromosomes; karyotyping and FISH.
Cytopenia	A reduction in the number of a type of blood cell.
Differentiation syndrome	A potentially severe inflammatory reaction occurring when leukemic cells are induced to mature by ATRA, arsenic or IDH inhibitors.
Disseminated intravascular coagulation (DIC)	Widespread activation of clotting that consumes clotting factors and platelets, causing bleeding.
Flow cytometry	Laser-based analysis of the proteins on and inside cells; used for immunophenotyping.
Graft-versus-host disease (GVHD)	Attack by donor immune cells on the recipient's tissues after allogeneic transplant.
Graft-versus-leukemia effect	Attack by donor immune cells on residual leukemia; the main curative mechanism of allogeneic transplant.
Haematopoiesis	The production of blood cells.

Term	Meaning
Hypomethylating agent	A drug (azacitidine, decitabine) that reverses abnormal DNA methylation in malignant cells.
IGHV mutation status	Whether the immunoglobulin heavy-chain gene in CLL cells has undergone somatic hypermutation; mutated is favourable.
Immunophenotype	The pattern of surface and intracellular markers on a cell.
Induction	The first phase of treatment, aimed at achieving remission.
Intrathecal	Given by injection into the cerebrospinal fluid via lumbar puncture.
Karyotype	The complete set of chromosomes in a cell, as visualised under the microscope.
Leukostasis	Obstruction of small blood vessels by very high numbers of white cells.
Lymphoblast	An immature lymphoid cell; the malignant cell of ALL.
Maintenance	Prolonged low-intensity treatment after consolidation to prevent relapse.
Measurable (minimal) residual disease (MRD)	Leukemic cells persisting after treatment below the level detectable by microscopy.
Myeloblast	An immature myeloid cell; the malignant cell of AML.
Myelodysplastic syndrome (MDS)	A clonal marrow disorder with ineffective haematopoiesis that may evolve into AML.
Neutropenia	A shortage of neutrophils, causing susceptibility to infection.
Next-generation sequencing (NGS)	High-throughput DNA sequencing used to detect mutations across many genes at once.
Philadelphia chromosome	The shortened chromosome 22 resulting from t(9;22); carries BCR::ABL1.
Refractory	Disease that does not respond to treatment.
Relapse	Return of disease after remission.
Richter transformation	Transformation of CLL into an aggressive lymphoma.
Splenomegaly	Enlargement of the spleen.
Thrombocytopenia	A shortage of platelets, causing bleeding and bruising.
Translocation	Exchange of material between two chromosomes, often creating a fusion gene.
Treatment-free remission	Sustained remission after deliberately stopping therapy, a goal in CML.
Tumour lysis syndrome	Metabolic disturbance from rapid breakdown of large numbers of cancer cells.
Tyrosine kinase inhibitor (TKI)	A drug that blocks a signalling enzyme; used for CML, Ph-positive ALL and FLT3-mutated AML.
Watch and wait	Active monitoring without treatment of early-stage, asymptomatic chronic leukemia.